INCH-POUND
MIL-DTL-24231/21E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/21D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, PLUG, TYPE I, MOLDED, 80-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 4.

2. If the plug shell is machined in the unaged condition, it shall be heat-treated to a hardness within the range of 250 to 320 Brinell or 24 to 34 Rockwell C measured in the ball bearing groove area.

3. Each coupling shall be examined for cracks in accordance with the liquid penetrant method of T9074-AS-GIB-010/271. Finding of cracks shall be cause for rejection.

4. Hydrostatic pressure test specified in MIL-DTL-24231 for mated plug and receptacle assembly (first article inspection) and for Type I molded plugs (conformance inspection) shall be reduced to 1,000 lbf/in^2 for 24 hours and 1 hour, respectively.

5. Ni-Cu plug sleeves shall be coated with plasma spray non-conductive material in accordance with S9320-AM-PRO-030/MLDG.

6. All dimensions are in inches.

7. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½ degree.

8. These connectors are for maintenance purposes only.

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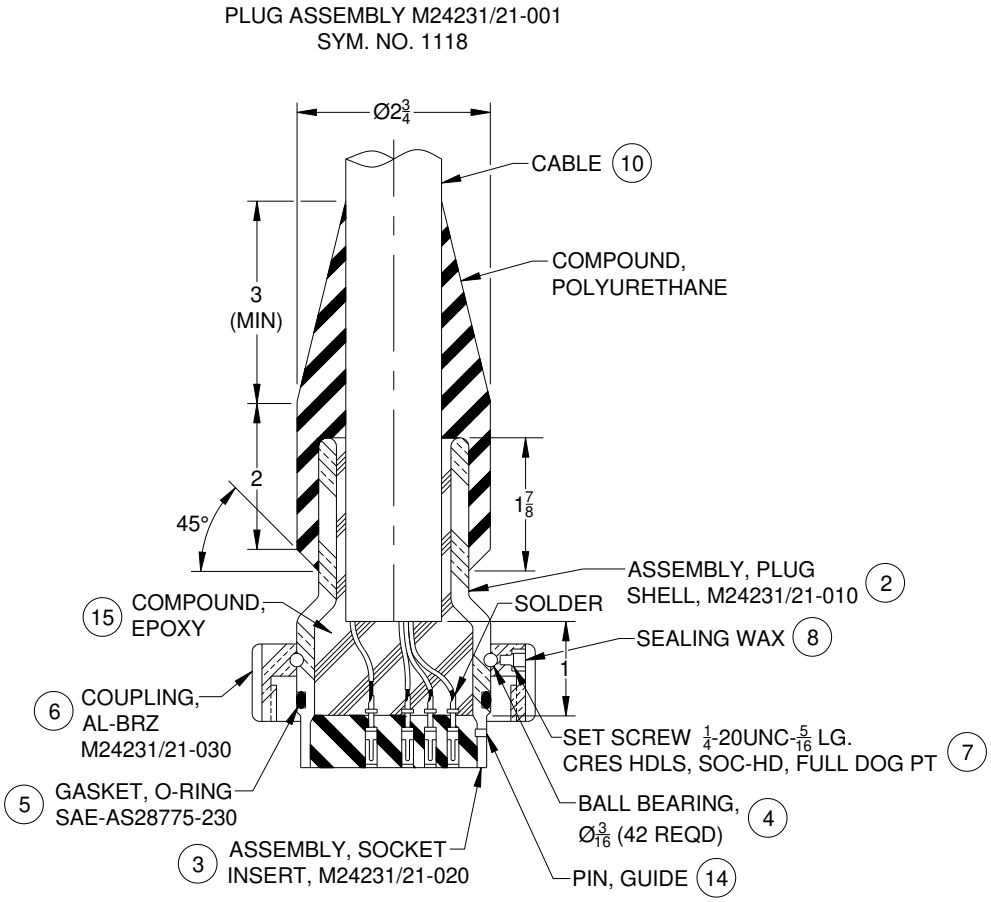
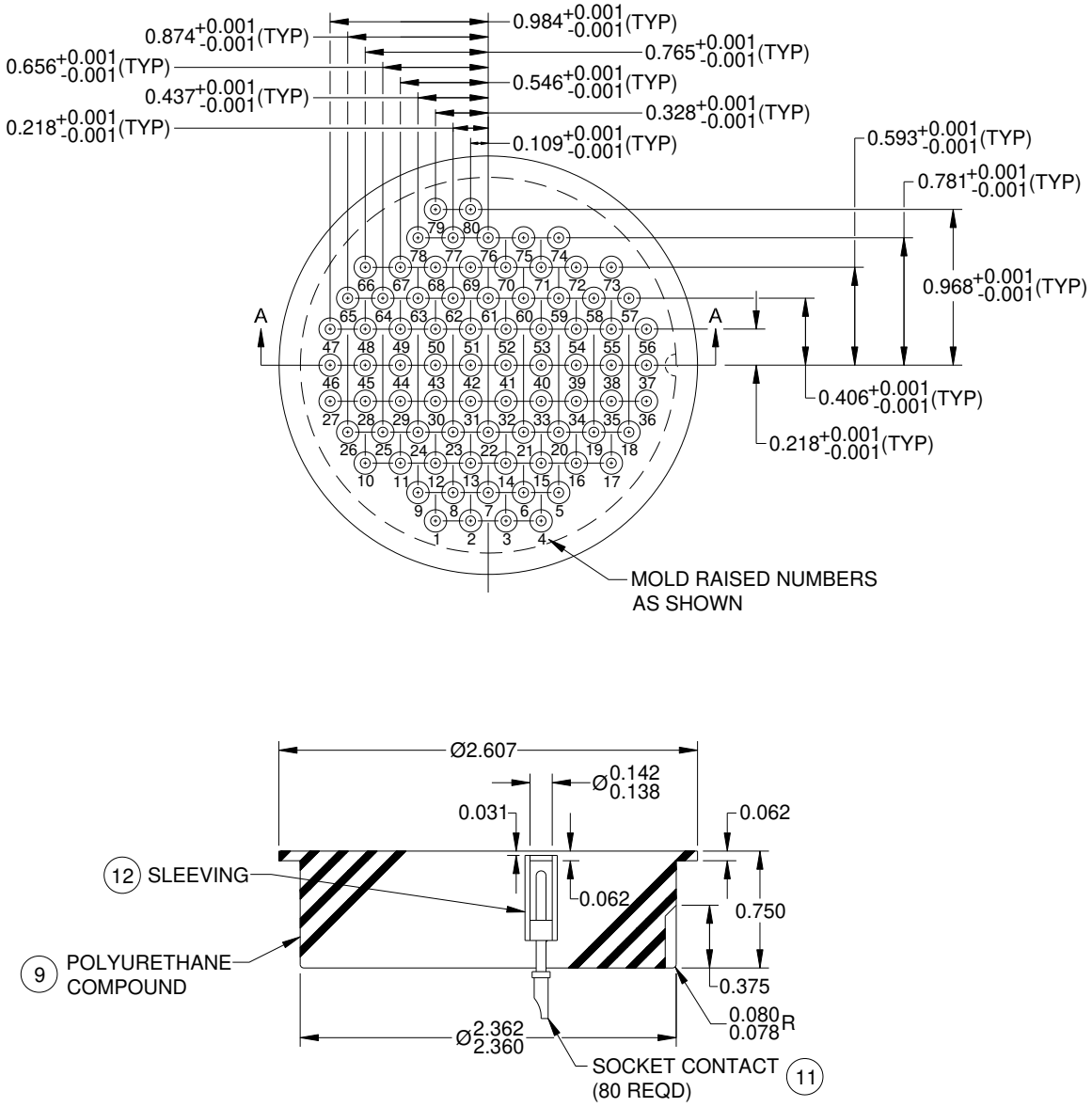


FIGURE 1. Connector, plug, 80-conductor.

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SECTION A-A

FIGURE 2. Assembly, socket insert – M24231/21-020. (3)

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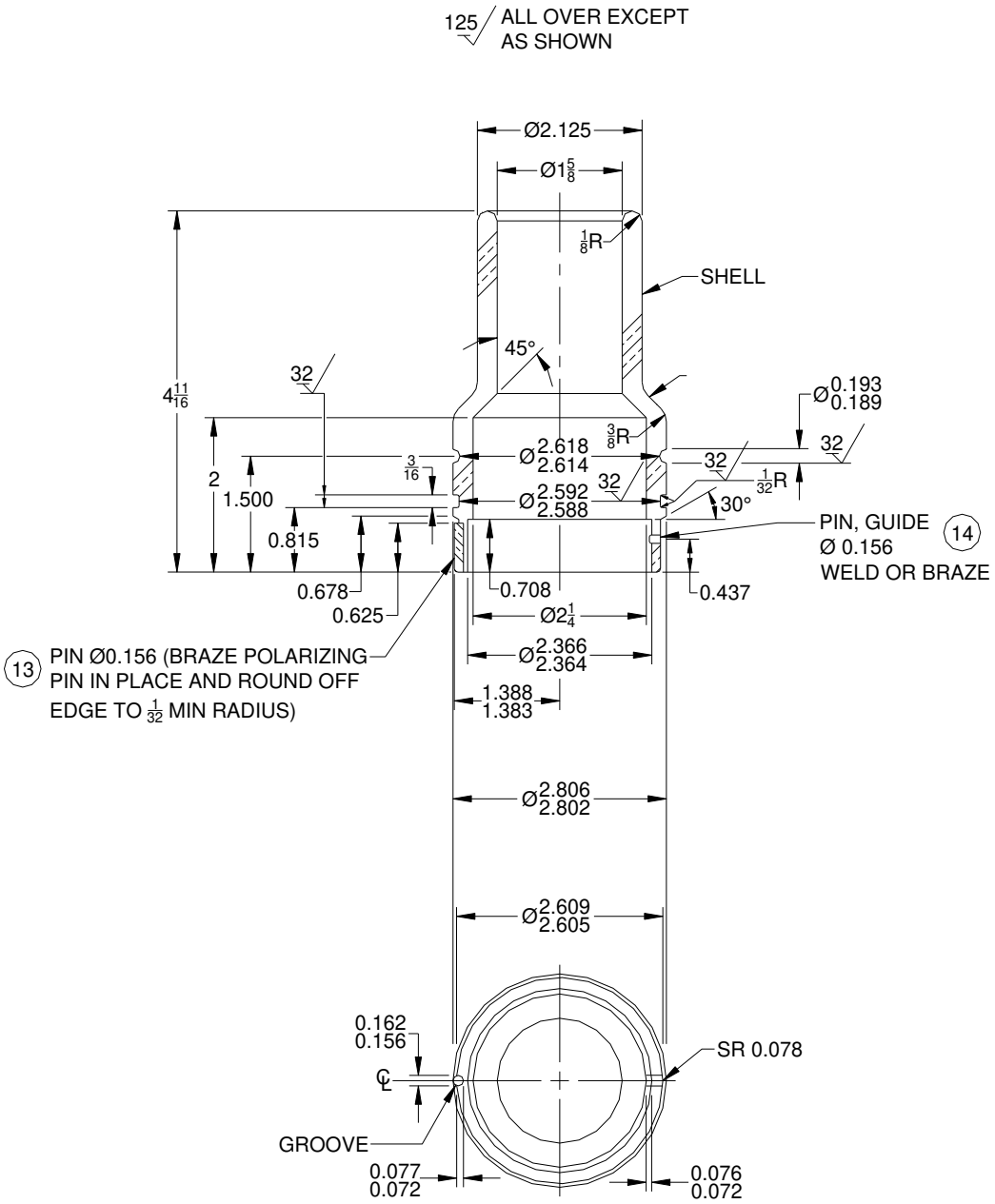


FIGURE 3. Assembly plug shell – M24231/21-010. (2)

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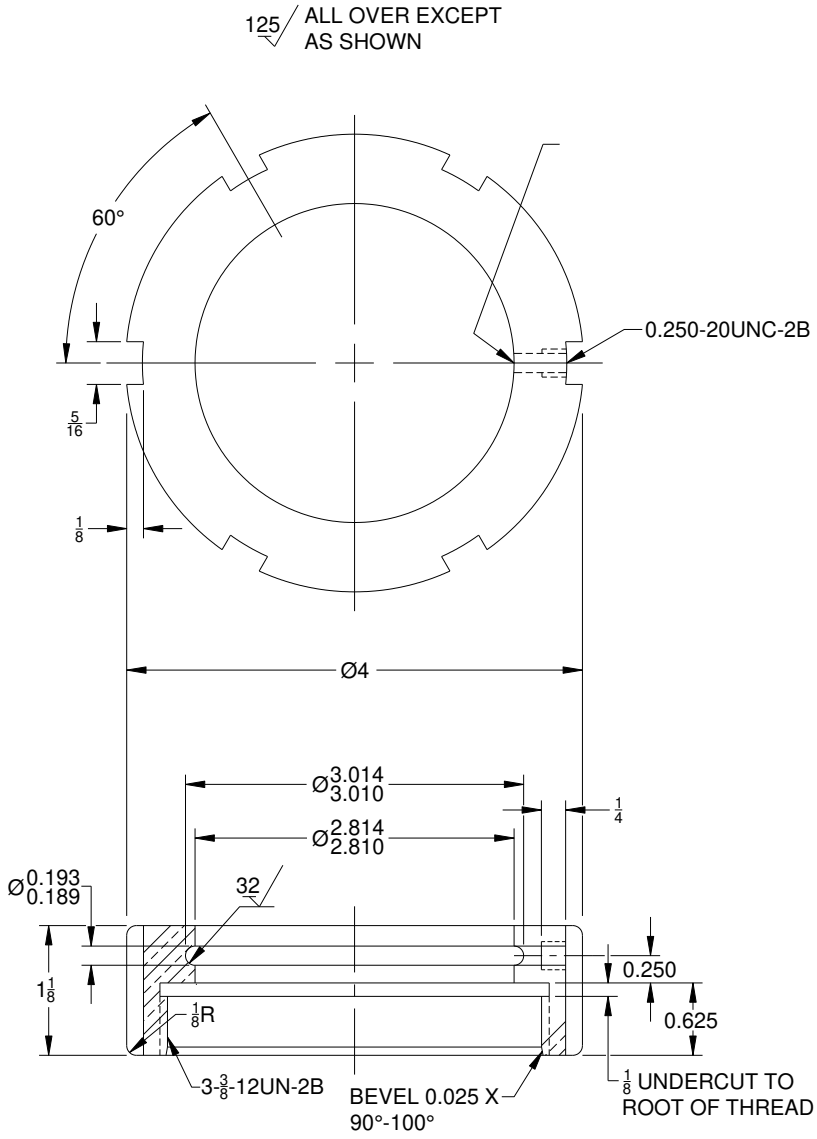


FIGURE 4. Coupling – M24231/21-030. (6)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/21- 3/	Material/specification	Remarks
		001		
1	Plug, 80-conductor, symbol no. 1118	1	------	See figure 1.
2	Shell, plug	1	Ni-Cu-Al, QQ-N-286	See figure 1 and figure 3.
3	Insert assembly, plug	1	See items 09, 11, and 12.	See figure 1 and figure 2.
4	Bearing, ball 3/16 diameter	42	Bronze, ASTM F2215	See figure 1.
5	Gasket, O-ring	1	SAE-AS28775-230 4/	Lubricate with silicone compound, SAE-AS8660; see figure 1.
6	Coupling	1	Ni-Al-BRZ, ASTM B150/B150M	See figure 4.
7	Screw, set, ¼-20UNC by 5/16 Lg, HDLS-Soc-Hd, full dog point	1	CRES, coml	See figure 1.
8	Wax, sealing	As required	Coml	See figure 1.
9	Compound, polyurethane	As required	5/	See figure 1 and figure 2.
10	Cable, Type ECM	As required	MIL-DTL-24643/38	See figure 1.
11	Contact, socket	80	MIL-DTL-24231	See figure 2, section A-A.
12	Sleeving, socket no. 9, 0.118 id by 9/16 long	As required	Type F, Subform U_a, Grade a, Class II, Category 1, color clear, MIL-I-631	See figure 2, section A-A.
13	Pin, polarizing, 0.156 diameter	1	Ni-Cu, Class A, QQ-N-281	See figure 3.
14	Pin, polarizing, 0.156 diameter, length to suit	1	Ni-Cu, Class A, QQ-N-281	See figure 3.
15	Compound, epoxy resin	As required	Cast epoxy resin compound 6/	Base EE4183 or RE2039, hardener HD-3561; see figure 1.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

5/ Molding compound to be procured in accordance with S9320-AM-PRO-020.

6/ Hysol base EE4183 or RE2039, hardener HD-3561, Hysol Division, Dexter Corporation, Olean, NY, CAGE 04347, or equal.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-070)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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